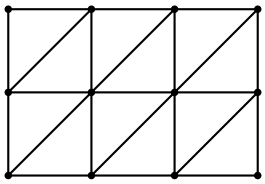
Mechanics of Flexible Structures & Soft Robots
MAE-263F
Fall 2026

MAE-263F Sample Midterm 2: Solutions

This sample midterm has 31 problems. The actual midterm will have 20 problems, each worth 0.5 pts.

Question 1

A rectangular plate with 4×3 nodes is meshed with 12 triangles as shown below (each square is cut by the diagonal from its lower-left to its upper-right corner).



How many edges and how many hinges does the mesh have?

(a) 23 edges and 23 hinges

(b) 17 edges and 7 hinges

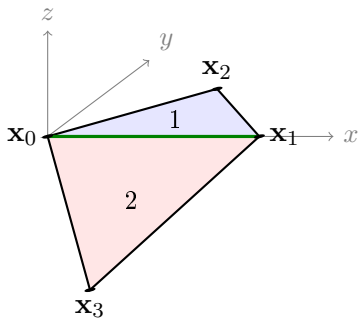
(c) **23 edges and 13 hinges** ✓

(d) 26 edges and 13 hinges

Answer: (c)
Solution. Edges: $3 \times 3 = 9$ horizontal, $4 \times 2 = 8$ vertical, and 6 diagonal, so 23 edges. A hinge is an edge shared by two triangles; only the $2(3+2) = 10$ boundary edges are not. Hinges: $23 - 10 = 13$.

Question 2

Consider the following hinge. The hinge has four nodes (in meters): $\mathbf{x}_0 = [0, 0, 0]$, $\mathbf{x}_1 = [2, 0, 0]$, $\mathbf{x}_2 = [1, 1, 0]$, and $\mathbf{x}_3 = [1, -1, -1]$, shown below. Write each node as $\mathbf{x}_k = [x_k, y_k, z_k]$. The hinge edge is $\mathbf{x}_0$–$\mathbf{x}_1$; triangle 1 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$ and triangle 2 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_3)$. The triangle normals are $\mathbf{n}_0 = (\mathbf{x}_1 - \mathbf{x}_0) \times (\mathbf{x}_2 - \mathbf{x}_0)$ and $\mathbf{n}_1 = (\mathbf{x}_3 - \mathbf{x}_0) \times (\mathbf{x}_1 - \mathbf{x}_0)$.



Compute the triangle normals $\mathbf{n}_0$ and $\mathbf{n}_1$.

(a) $\mathbf{n}_0 = [0, 0, 2]$, $\mathbf{n}_1 = [0, 2, -2]$

(b) $\mathbf{n_0 = [0, 0, 2]}$, $\mathbf{n_1 = [0, -2, 2]}$ ✓

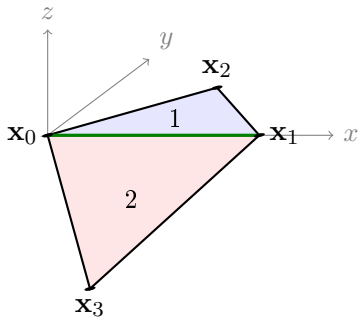
(c) $\mathbf{n}_0 = [0, 0, 1]$, $\mathbf{n}_1 = [0, -1, -1]$

(d) $\mathbf{n}_0 = [2, 0, 0]$, $\mathbf{n}_1 = [0, -2, 2]$

Answer: (b)
Solution. $\mathbf{n}_0 = [2, 0, 0] \times [1, 1, 0] = [0 \cdot 0 - 0 \cdot 1,\ 0 \cdot 1 - 2 \cdot 0,\ 2 \cdot 1 - 0 \cdot 1] = [0, 0, 2]$.
$\mathbf{n}_1 = [1, -1, -1] \times [2, 0, 0] = [(-1)(0) - (-1)(0),\ (-1)(2) - (1)(0),\ (1)(0) - (-1)(2)] = [0, -2, 2]$.
(Option (a) computes $\mathbf{e}^0 \times (\mathbf{x}_3 - \mathbf{x}_0)$ instead.)

Question 3

Consider the same hinge as in Question 2 (repeated here). The hinge has four nodes (in meters): $\mathbf{x}_0 = [0, 0, 0]$, $\mathbf{x}_1 = [2, 0, 0]$, $\mathbf{x}_2 = [1, 1, 0]$, and $\mathbf{x}_3 = [1, -1, -1]$, shown below. Write each node as $\mathbf{x}_k = [x_k, y_k, z_k]$. The hinge edge is $\mathbf{x}_0$–$\mathbf{x}_1$; triangle 1 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$ and triangle 2 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_3)$. The triangle normals are $\mathbf{n}_0 = (\mathbf{x}_1 - \mathbf{x}_0) \times (\mathbf{x}_2 - \mathbf{x}_0)$ and $\mathbf{n}_1 = (\mathbf{x}_3 - \mathbf{x}_0) \times (\mathbf{x}_1 - \mathbf{x}_0)$.



The hinge angle θ is the signed angle from $\mathbf{n}_0$ to $\mathbf{n}_1$ about the hinge vector $\mathbf{e}^0 = \mathbf{x}_1 - \mathbf{x}_0$: it is positive if $\mathbf{n}_0 \times \mathbf{n}_1$ points along $\mathbf{e}^0$ and negative otherwise. What is θ?

(a) $-\pi/4$

(b) $\pi/2$

(c) $3\pi/4$

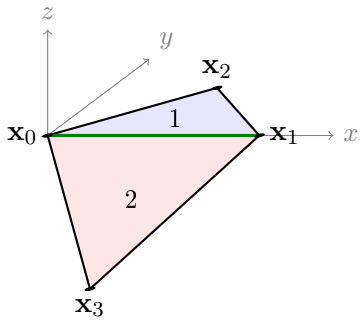
(d) $\boldsymbol{\pi/4}$ ✓

Answer: (d)
Solution. From Question 2, $\mathbf{n}_0 = [0, 0, 2]$ and $\mathbf{n}_1 = [0, -2, 2]$. $\cos\theta = \frac{\mathbf{n}_0 \cdot \mathbf{n}_1}{|\mathbf{n}_0||\mathbf{n}_1|} = \frac{4}{2 \cdot 2\sqrt{2}} = \frac{1}{\sqrt{2}}$, so $|\theta| = \pi/4$. Sign: $\mathbf{n}_0 \times \mathbf{n}_1 = [4, 0, 0]$, which points along $\mathbf{e}^0 = [2, 0, 0]$. So $\theta = +\pi/4$.

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Question 4

Consider the same hinge as in Questions 2–3 (repeated here), whose hinge angle is $\theta = \pi/4$. The hinge has four nodes (in meters): $\mathbf{x}_0 = [0, 0, 0]$, $\mathbf{x}_1 = [2, 0, 0]$, $\mathbf{x}_2 = [1, 1, 0]$, and $\mathbf{x}_3 = [1, -1, -1]$, shown below. Write each node as $\mathbf{x}_k = [x_k, y_k, z_k]$. The hinge edge is $\mathbf{x}_0$–$\mathbf{x}_1$; triangle 1 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$ and triangle 2 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_3)$. The triangle normals are $\mathbf{n}_0 = (\mathbf{x}_1 - \mathbf{x}_0) \times (\mathbf{x}_2 - \mathbf{x}_0)$ and $\mathbf{n}_1 = (\mathbf{x}_3 - \mathbf{x}_0) \times (\mathbf{x}_1 - \mathbf{x}_0)$.



The bending stiffness is $k_b = 2\,\mathrm{N\,m}$ and the bending energy is $E_b = \frac{1}{2} k_b (\theta - \bar{\theta})^2$. What is E_b if the hinge is a *plate* ($\bar{\theta} = 0$), and if it is a *shell* with $\bar{\theta} = \pi/6$?

(a) Plate: $\pi^2/8\,\mathrm{J}$; shell: $\pi^2/72\,\mathrm{J}$

(b) Plate: $\pi^2/16\,\mathrm{J}$; shell: $25\pi^2/144\,\mathrm{J}$

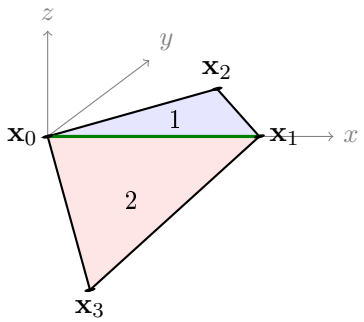
(c) **Plate: $\boldsymbol{\pi^2/16\,\mathrm{J}}$; shell: $\boldsymbol{\pi^2/144\,\mathrm{J}}$** ✓

(d) Plate: $\pi/4\,\mathrm{J}$; shell: $\pi/12\,\mathrm{J}$

Answer: (c)
Solution. Plate: $E_b = \frac{1}{2} \cdot 2 \cdot (\pi/4)^2 = \pi^2/16\,\mathrm{J}$. Shell: $E_b = \frac{1}{2} \cdot 2 \cdot (\pi/4 - \pi/6)^2 = (\pi/12)^2 = \pi^2/144\,\mathrm{J}$. (Option (b) uses $\theta + \bar{\theta} = 5\pi/12$.)

Question 5

Consider a setup similar to Question 4 (repeated here): a *plate* ($\bar{\theta} = 0$) with $k_b = 2\,\mathrm{N\,m}$ and $E_b = \frac{1}{2} k_b \theta^2$. The hinge has four nodes (in meters): $\mathbf{x}_0 = [0, 0, 0]$, $\mathbf{x}_1 = [2, 0, 0]$, $\mathbf{x}_2 = [1, 1, 0]$, and $\mathbf{x}_3 = [1, -1, -1]$, shown below. Write each node as $\mathbf{x}_k = [x_k, y_k, z_k]$. The hinge edge is $\mathbf{x}_0$–$\mathbf{x}_1$; triangle 1 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$ and triangle 2 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_3)$. The triangle normals are $\mathbf{n}_0 = (\mathbf{x}_1 - \mathbf{x}_0) \times (\mathbf{x}_2 - \mathbf{x}_0)$ and $\mathbf{n}_1 = (\mathbf{x}_3 - \mathbf{x}_0) \times (\mathbf{x}_1 - \mathbf{x}_0)$.



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Node $\mathbf{x}_3$ is now moved from $[1,-1,-1]$ to $[1,0,-1]$, and all other nodes stay where they are. What is the new bending energy E_b?

(a) $\boldsymbol{\pi^2/4\,\mathrm{J}}$ ✓

(b) 0

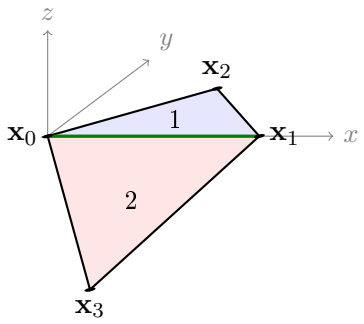
(c) $\pi^2/16\,\mathrm{J}$

(d) $\pi^2/2\,\mathrm{J}$

Answer: (a)
Solution. $\mathbf{n}_1 = [1,0,-1]\times[2,0,0] = [0,-2,0]$, which is perpendicular to $\mathbf{n}_0 = [0,0,2]$; $\mathbf{n}_0 \times \mathbf{n}_1 = [4,0,0]$ points along $\mathbf{e}^0$, so $\theta = \pi/2$. $E_b = \frac{1}{2}\cdot 2\cdot(\pi/2)^2 = \pi^2/4\,\mathrm{J}$.

Question 6

Consider the same hinge as in Questions 2–4 (repeated here), as a plate with $E_b = \frac{1}{2}k_b\theta^2$. The hinge has four nodes (in meters): $\mathbf{x}_0 = [0,0,0]$, $\mathbf{x}_1 = [2,0,0]$, $\mathbf{x}_2 = [1,1,0]$, and $\mathbf{x}_3 = [1,-1,-1]$, shown below. Write each node as $\mathbf{x}_k = [x_k, y_k, z_k]$. The hinge edge is $\mathbf{x}_0$–$\mathbf{x}_1$; triangle 1 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$ and triangle 2 is $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_3)$. The triangle normals are $\mathbf{n}_0 = (\mathbf{x}_1 - \mathbf{x}_0)\times(\mathbf{x}_2 - \mathbf{x}_0)$ and $\mathbf{n}_1 = (\mathbf{x}_3 - \mathbf{x}_0)\times(\mathbf{x}_1 - \mathbf{x}_0)$.



Which component of the gradient of the bending energy is **zero**?

(a) $\partial E_b/\partial y_3$

(b) $\partial E_b/\partial z_0$

(c) $\partial E_b/\partial z_2$

(d) $\boldsymbol{\partial E_b/\partial y_2}$ ✓

Answer: (d)
Solution. Triangle 1 lies in the plane $z = 0$. Moving $\mathbf{x}_2$ in y keeps it in that plane, so $\mathbf{n}_0$ keeps its direction; $\mathbf{n}_1$ does not depend on $\mathbf{x}_2$. Thus θ, and therefore E_b, do not change: $\partial E_b/\partial y_2 = 0$. Moving $\mathbf{x}_3$ in y, $\mathbf{x}_0$ in z, or $\mathbf{x}_2$ in z tilts a triangle and changes θ, so those components are nonzero. (From `gradEb_hessEb_Shell`: $\partial E_b/\partial y_3 = \pi/4$, $\partial E_b/\partial z_0 = 3\pi/8$, $\partial E_b/\partial z_2 = -\pi/2$.)

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Question 7

Let $\mathbf{f}(x,y) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} x^2 + 3y \\ xy^2 \end{bmatrix}$. At $(x,y) = (1,1)$, estimate the Jacobian $J_{ij} = \partial f_i / \partial q_j$ (with $q_1 = x$, $q_2 = y$) using the forward finite difference

$$\frac{\partial f_i}{\partial q_j} \approx \frac{f_i(\mathbf{q} + \Delta q\, \mathbf{e}_j) - f_i(\mathbf{q})}{\Delta q},$$

where $\mathbf{e}_j$ perturbs only q_j, with $\Delta q = 0.1$. That is, perturb x to 1.1 (keeping $y = 1$) for the first column, and perturb y to 1.1 (keeping $x = 1$) for the second column. What is the finite-difference estimate of $\mathbf{J}$?

(a) $\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 2.1 & 1 \\ 3 & 2.1 \end{bmatrix}$

(c) $\begin{bmatrix} 0.21 & 0.3 \\ 0.1 & 0.21 \end{bmatrix}$

(d) $\begin{bmatrix} \mathbf{2.1} & \mathbf{3} \\ \mathbf{1} & \mathbf{2.1} \end{bmatrix}$ ✓

Answer: (d)
Solution. $\mathbf{f}(1,1) = [4,1]^T$. Perturbing x: $\mathbf{f}(1.1,1) = [1.21 + 3,\ 1.1]^T = [4.21, 1.1]^T$, so the first column is $([4.21,1.1]^T - [4,1]^T)/0.1 = [2.1,1]^T$. Perturbing y: $\mathbf{f}(1,1.1) = [1 + 3.3,\ 1.21]^T = [4.3, 1.21]^T$, so the second column is $[3,2.1]^T$. The exact Jacobian is $\begin{bmatrix} 2x & 3 \\ y^2 & 2xy \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ (option (a)); the finite difference is off by 0.1 in the entries that depend on the perturbed variable.

Question 8

An edge of a plate has undeformed length $l = 0.1\,\text{m}$. The plate has Young's modulus $Y = 1\,\text{MPa}$ and thickness $h = 1\,\text{mm}$, so $k_s = \frac{\sqrt{3}}{2} Y h l^2 = 5\sqrt{3}\,\text{N m}$. The edge is stretched to a length of $0.11\,\text{m}$. What is its stretching energy $E_s = \frac{1}{2} k_s \left(\frac{\|\mathbf{e}\|}{l} - 1 \right)^2$?

(a) $\sqrt{3}/20\,\text{J}$

(b) $\mathbf{\sqrt{3}/40\,J}$ ✓

(c) $\sqrt{3}/4\,\text{J}$

(d) $\sqrt{3}/4000\,\text{J}$

Answer: (b)
Solution. Strain: $0.11/0.1 - 1 = 0.1$. $E_s = \frac{1}{2} \cdot 5\sqrt{3} \cdot 0.1^2 = \frac{5\sqrt{3}}{200} = \frac{\sqrt{3}}{40}\,\text{J}$. (Option (d) uses the change in length, 0.01 m, as if it were the strain.)

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Question 9

The thickness h of a plate is doubled. By what factors do the bending stiffness $k_b = \frac{2}{\sqrt{3}}\frac{Yh^3}{12}$ and the stretching stiffness $k_s = \frac{\sqrt{3}}{2}Yhl^2$ change?

(a) Both double.

(b) Both increase by a factor of 8.

(c) k_b doubles; k_s increases by a factor of 8.

(d) **k_b increases by a factor of 8; k_s doubles.** ✓

Answer: (d)
Solution. $k_b \propto h^3$, so it increases by $2^3 = 8$; $k_s \propto h$, so it doubles. This is why thin plates and shells bend easily but resist stretching.

Question 10

A discrete plate has N nodes. How many DOFs does it have, and what are the sizes of the gradient vector of one hinge's bending energy and of one edge's stretching energy?

(a) **$3N$ DOFs; hinge gradient: 12; edge gradient: 6** ✓

(b) $4N - 1$ DOFs; hinge gradient: 12; edge gradient: 6

(c) $3N$ DOFs; hinge gradient: 9; edge gradient: 6

(d) $3N$ DOFs; hinge gradient: 12; edge gradient: 3

Answer: (a)
Solution. Each node has three coordinates and there are no twist angles, so the DOF vector has $3N$ entries ($4N - 1$ is the DER rod). The bending energy of a hinge depends on four nodes (12 DOFs), and the stretching energy of an edge on two nodes (6 DOFs); `gradEb_hessEb_Shell` returns a 12-vector and a 12×12 Hessian.

Question 11

In a discrete shell simulation, how is the natural hinge angle $\bar{\theta}$ of each hinge obtained?

(a) It is recomputed at every time step from the current shape.

(b) **It is computed once, from the initial (undeformed) shape, before time stepping starts; for a plate, it is zero.** ✓

(c) It is zero for every hinge of a shell.

(d) It is a degree of freedom solved for by Newton's method.

Answer: (b)
Solution. The natural hinge angles $\bar{\theta}_k$ (and the undeformed edge lengths) are computed once from the undeformed configuration, usually $\mathbf{q}(0)$, and stored before time stepping. For a plate, the undeformed shape is flat, so $\bar{\theta}_k = 0$; for a shell, $\bar{\theta}_k \neq 0$ in general.

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Question 12

A discrete shell has four nodes, $\mathbf{x}_0$, $\mathbf{x}_1$, $\mathbf{x}_2$, $\mathbf{x}_3$, each with three coordinates. Its DOF vector is $\mathbf{q} = [\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3]$ (size 12, Python indices 0–11), so node $\mathbf{x}_k$ occupies `q[3*k : 3*k+3]`. Nodes $\mathbf{x}_0$, $\mathbf{x}_1$, $\mathbf{x}_2$ are clamped (fixed) and $\mathbf{x}_3$ is free. Which Python indices of $\mathbf{q}$ are the free DOFs?

(a) `0, 1, 2`

(b) `0` through `8`

(c) `9, 10, 11` ✓

(d) All 12 indices

Answer: (c)
Solution. Node $\mathbf{x}_3$ occupies `q[9:12]`, so `freeIndex = np.arange(9, 12)`, i.e., indices 9, 10, 11. Only these rows and columns of $\mathbf{f}$ and $\mathbf{J}$ are solved.

Question 13

In resistive force theory, the drag on a rod depends on the direction of motion: moving along the rod is resisted less than moving sideways. So the velocity $\mathbf{v}$ (a column vector) of a node is decomposed into a tangential velocity $\mathbf{v}_t$, along the unit tangent $\mathbf{t}$ (a column vector), and a normal velocity $\mathbf{v}_n$, perpendicular to $\mathbf{t}$, with $\mathbf{v} = \mathbf{v}_t + \mathbf{v}_n$. What is $\mathbf{v}_t$?

(a) $(\mathbf{t}^T\mathbf{t})\,\mathbf{v}$

(b) $(\mathbf{I} - \mathbf{t}\mathbf{t}^T)\,\mathbf{v}$

(c) $\mathbf{t} \times \mathbf{v}$

(d) $\boldsymbol{(\mathbf{t}\mathbf{t}^T)\,\mathbf{v}}$ ✓

Answer: (d)
Solution. $(\mathbf{t}\mathbf{t}^T)\mathbf{v} = \mathbf{t}(\mathbf{t}^T\mathbf{v}) = (\mathbf{t}\cdot\mathbf{v})\,\mathbf{t}$, the projection of $\mathbf{v}$ onto $\mathbf{t}$ ($\mathbf{P}_t = \mathbf{t}\mathbf{t}^T$, Eq. 10.7). Option (b) is the normal component; option (a) is just $\mathbf{v}$ because $\mathbf{t}^T\mathbf{t} = 1$.

Question 14

As in Question 13, the velocity $\mathbf{v}$ of a node of a rod is decomposed into a tangential velocity $\mathbf{v}_t$ (along the unit tangent $\mathbf{t}$) and a normal velocity $\mathbf{v}_n$ (perpendicular to $\mathbf{t}$), with $\mathbf{v} = \mathbf{v}_t + \mathbf{v}_n$. For $\mathbf{t} = [0.6, 0.8, 0]^T$ and $\mathbf{v} = [1, 2, 0]^T$ m/s, what is the normal velocity $\mathbf{v}_n$?

(a) $[1.32, 1.76, 0]^T$ m/s

(b) $\boldsymbol{[-0.32, 0.24, 0]^T}$ **m/s** ✓

(c) $[0.32, -0.24, 0]^T$ m/s

(d) $[0.4, 1.2, 0]^T$ m/s

Answer: (b)
Solution. $\mathbf{v}_n = (\mathbf{I} - \mathbf{t}\mathbf{t}^T)\,\mathbf{v} = \mathbf{v} - (\mathbf{t}\cdot\mathbf{v})\,\mathbf{t}$. Here $\mathbf{t}\cdot\mathbf{v} = 0.6 + 1.6 = 2.2$, so $\mathbf{v}_t = 2.2\,\mathbf{t} = [1.32, 1.76, 0]^T$ and $\mathbf{v}_n = [1, 2, 0]^T - [1.32, 1.76, 0]^T = [-0.32, 0.24, 0]^T$. Check: it is perpendicular to $\mathbf{t}$ ($-0.192 + 0.192 = 0$).

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Question 15

In resistive force theory, the drag on node k is $\mathbf{f}_k = -l_k\,(C_t\mathbf{P}_t + C_p\mathbf{P}_n)\,\mathbf{u}_k$, with $\mathbf{P}_t = \mathbf{t}\mathbf{t}^T$ and $\mathbf{P}_n = \mathbf{I} - \mathbf{P}_t$. Let $l_k = 0.01\,\text{m}$, $C_t = 1$, $C_p = 2$ (in $\text{Pa}\cdot\text{s}$), $\mathbf{t} = [1,0,0]^T$, and $\mathbf{u}_k = [0.1, 0.1, 0]^T\,\text{m/s}$ (no background flow). What is $\mathbf{f}_k$?

(a) $\mathbf{[-0.001, -0.002, 0]^T\,N}$ ✓

(b) $[-0.002, -0.001, 0]^T\,\text{N}$

(c) $[0.001, 0.002, 0]^T\,\text{N}$

(d) $[-0.1, -0.2, 0]^T\,\text{N}$

Answer: (a)
Solution. Tangential part of $\mathbf{u}_k$: $[0.1, 0, 0]^T$; normal part: $[0, 0.1, 0]^T$. $\mathbf{f}_k = -0.01\,(1\cdot[0.1,0,0]^T + 2\cdot[0,0.1,0]^T) = [-0.001, -0.002, 0]^T\,\text{N}$. Moving sideways costs twice as much drag as moving along the rod.

Question 16

At high Reynolds number, the drag on node k of a rod is $\mathbf{f}_k = -k_k\|\mathbf{u}_k^{\text{rel}}\|\,\mathbf{u}_k^{\text{rel}}$. Here $\mathbf{u}_k^{\text{rel}} = \mathbf{u}_k - \mathbf{U}_\infty$ is the velocity of the node relative to the background flow, and $k_k = \frac{1}{2}\rho C_D A_k$ is a constant that depends on the fluid density ρ, the drag coefficient C_D, and the projected area A_k of the rod segment around node k. If the relative velocity is tripled (same direction), by what factor does the magnitude of the drag force change?

(a) 3

(b) 6

(c) **9** ✓

(d) 27

Answer: (c)
Solution. $\|\mathbf{f}_k\| = k_k\|\mathbf{u}_k^{\text{rel}}\|^2$, so tripling the speed multiplies the drag by $3^2 = 9$. (In the low-Re model the drag is linear, so it would only triple.)

Question 17

A rod of diameter $L = 0.01\,\text{m}$ moves at $v = 1\,\text{m/s}$ in water ($\rho = 1000\,\text{kg/m}^3$, $\mu = 10^{-3}\,\text{Pa}\cdot\text{s}$). What is the Reynolds number $Re = \rho v L/\mu$, and which hydrodynamic model is appropriate?

(a) $Re = 10$; resistive force theory (low Re)

(b) $Re = 10^4$; resistive force theory (low Re)

(c) $Re = 10^{-4}$; quadratic drag (high Re)

(d) $\boldsymbol{Re = 10^4}$**; quadratic drag (high Re)** ✓

Answer: (d)
Solution. $Re = 1000\cdot 1\cdot 0.01/10^{-3} = 10^4 \gg 1$: inertial effects dominate, so the quadratic (high-Re) drag model is appropriate.

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Question 18

A rod has nodes $\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{N-1}$. The two end nodes, $\mathbf{x}_0$ and $\mathbf{x}_{N-1}$, belong to only one edge each; all other nodes are *interior* nodes and belong to two edges (the edge before and the edge after). The tangent at an interior node $\mathbf{x}_k$ is the normalized average of the unit tangents of these two edges. For an interior node with $\mathbf{x}_k - \mathbf{x}_{k-1} = [2, 0, 0]$ and $\mathbf{x}_{k+1} - \mathbf{x}_k = [0, 3, 0]$, what is the node tangent?

(a) $\mathbf{\frac{1}{\sqrt{2}}[1, 1, 0]}$ ✓

(b) $\frac{1}{\sqrt{13}}[2, 3, 0]$

(c) $[1, 1, 0]$

(d) $[0, 1, 0]$

Answer: (a)
Solution. Unit edge tangents: $[1, 0, 0]$ and $[0, 1, 0]$. Their sum is $[1, 1, 0]$, and normalizing gives $\frac{1}{\sqrt{2}}[1, 1, 0]$. Option (b) adds the edges before normalizing them, which weights the longer edge more.

Question 19

In the predictor–corrector method, a node is held fixed on the ground. After the solve, its reaction force points *toward* the ground, i.e., the ground would have to pull the node down to keep it there. What should the algorithm do?

(a) Keep the node fixed, because it is already on the ground.

(b) **Release the node (make its vertical DOF free again), because the ground can only push, not pull, and then do a corrector solve.** ✓

(c) Move the node below the ground to balance the force.

(d) Nothing; reaction forces do not affect the contact constraints.

Answer: (b)
Solution. Ground contact can only push. A reaction that pulls the node toward the ground would be an adhesive force, so the constraint is released (Eq. 11.11). Because the set of constraints changed, a corrector solve is needed.

Question 20

Two masses hang in a chain: spring $k_1 = 30\,\mathrm{N/m}$ connects a fixed support to mass $m_1 = 1\,\mathrm{kg}$, and spring $k_2 = 30\,\mathrm{N/m}$ connects m_1 to mass $m_2 = 2\,\mathrm{kg}$. Gravity is $g = 10\,\mathrm{m/s^2}$. At static equilibrium, what is the displacement of m_2 from its position when both springs are unstretched?

(a) $1.00\,\mathrm{m}$

(b) $2.00\,\mathrm{m}$

(c) **$1.67\,\mathrm{m}$** ✓

(d) $0.67\,\mathrm{m}$

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Answer: (c)
Solution. Spring 1 carries the weight of both masses, $(1+2)\cdot 10 = 30\,\text{N}$, so it stretches 1 m. Spring 2 carries only m_2: 20 N, so it stretches $2/3\,\text{m}$. The displacement of m_2 is $1 + 2/3 = 1.67\,\text{m}$.

Question 21

A mass m is attached to a spring of stiffness k (no damping, no gravity). Its position $x(t)$ obeys

$$m\ddot{x} + kx = 0.$$

Writing the velocity as $u = \dot{x}$, this is the first-order system $\dot{x} = u$ and $\dot{u} = a$, where $a = -kx/m$ is the acceleration. The *explicit* Euler method advances from time t to $t + \Delta t$ using only the known (old) values at time t:

$$x^{\text{new}} = x^{\text{old}} + \Delta t\, u^{\text{old}}, \qquad u^{\text{new}} = u^{\text{old}} + \Delta t\, a^{\text{old}}, \qquad a^{\text{old}} = -kx^{\text{old}}/m.$$

Let $m = 1\,\text{kg}$, $k = 100\,\text{N/m}$, and $\Delta t = 0.01\,\text{s}$. The mass starts at $x^{\text{old}} = 0.1\,\text{m}$ with $u^{\text{old}} = 0$. After one explicit Euler step, what are x^{new} and u^{new}?

(a) $x = 0.099\,\text{m}$, $u = -0.1\,\text{m/s}$

(b) $x = 0.1\,\text{m}$, $u = 0$

(c) $x = 0.09\,\text{m}$, $u = -1\,\text{m/s}$

(d) **$x = 0.1\,\text{m}$, $u = -0.1\,\text{m/s}$** ✓

Answer: (d)
Solution. $x^{\text{new}} = 0.1 + 0.01\cdot 0 = 0.1\,\text{m}$. $a^{\text{old}} = -100\cdot 0.1/1 = -10\,\text{m/s}^2$, so $u^{\text{new}} = 0 + 0.01\cdot(-10) = -0.1\,\text{m/s}$. Both updates use only old values.

Question 22

An undamped mass–spring system is simulated with explicit Euler (as in Question 21) with any time step Δt. What happens to the oscillation over a long simulation?

(a) **Its amplitude grows without bound.** ✓

(b) Its amplitude decays to zero.

(c) It matches the exact solution.

(d) The mass stays at its initial position.

Answer: (a)
Solution. Explicit Euler adds energy to an undamped oscillator at every step, so the amplitude grows without bound (HW1, Question 10). Implicit Euler does the opposite: it removes energy.

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Question 23

An undamped mass–spring system is simulated with implicit Euler, and its oscillation slowly dies out. What happens if the time step Δt is made smaller?

(a) The oscillation dies out faster.

(b) **The oscillation dies out more slowly, because the artificial damping is weaker.** ✓

(c) Nothing changes; the artificial damping does not depend on Δt.

(d) The simulation becomes unstable.

Answer: (b)
Solution. Implicit Euler's artificial (numerical) damping decreases as Δt decreases, so the oscillation persists longer and is closer to the exact (undamped) solution (HW1, Question 9).

Question 24

The Newmark–Beta method advances the position x, velocity $\dot{x}$, and acceleration $\ddot{x}$ from time step n to $n+1$ (step size Δt) using

$$x_{n+1} = x_n + \Delta t\, \dot{x}_n + \Delta t^2 \left[\left(\tfrac{1}{2} - \beta\right) \ddot{x}_n + \beta\, \ddot{x}_{n+1}\right], \qquad \dot{x}_{n+1} = \dot{x}_n + \Delta t \left[(1-\gamma)\, \ddot{x}_n + \gamma\, \ddot{x}_{n+1}\right],$$

where β and γ are parameters that weight the old acceleration $\ddot{x}_n$ against the new one $\ddot{x}_{n+1}$. Which choice of β and γ is unconditionally stable but *strongly dissipative* (the energy of an undamped oscillator decays, similar to implicit Euler)?

(a) $\beta = 0.25$, $\gamma = 0.5$

(b) $\beta = 0$, $\gamma = 0.5$

(c) $\boldsymbol{\beta = 0.5}$**, **$\boldsymbol{\gamma = 1.0}$ ✓

(d) All choices conserve energy.

Answer: (c)
Solution. $\beta = 0.5$, $\gamma = 1.0$ is the fully implicit Newmark method: unconditionally stable but strongly dissipative (Ch. 6.2). $\beta = 0.25$, $\gamma = 0.5$ (average acceleration) is time-symmetric and conserves energy for undamped linear systems.

Question 25

A metal sphere ($\rho_{\text{metal}} = 7000\,\text{kg/m}^3$) is released from rest in a fluid ($\rho_{\text{fluid}} = 1000\,\text{kg/m}^3$) with no viscosity. With $g = 9.8\,\text{m/s}^2$, what is its speed after 2 s?

(a) 19.6 m/s

(b) **16.8 m/s** ✓

(c) 2.8 m/s

(d) 8.4 m/s

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Answer: (b)
Solution. Net downward force $= \frac{4}{3}\pi R^3(\rho_{\text{metal}} - \rho_{\text{fluid}})g$ and mass $= \frac{4}{3}\pi R^3\rho_{\text{metal}}$, so $a = g\,(1 - 1000/7000) = 8.4\,\text{m/s}^2$ and $v = 8.4 \cdot 2 = 16.8\,\text{m/s}$.

Question 26

A metal sphere of radius $R = 0.01\,\text{m}$ ($\rho_{\text{metal}} = 10000\,\text{kg/m}^3$) falls in a viscous fluid ($\rho_{\text{fluid}} = 1000\,\text{kg/m}^3$, $\mu = 1000\,\text{Pa}\cdot\text{s}$) with $g = 10\,\text{m/s}^2$. The drag is $6\pi\mu Rv$. What is the terminal speed?

(a) **2×10^{-3} m/s** ✓

(b) $2.2 \times 10^{-3}\,\text{m/s}$

(c) $9 \times 10^{-3}\,\text{m/s}$

(d) $2 \times 10^{-1}\,\text{m/s}$

Answer: (a)
Solution. At terminal speed, $6\pi\mu Rv = \frac{4}{3}\pi R^3(\rho_{\text{metal}} - \rho_{\text{fluid}})g$, so $v = \frac{2}{9}\frac{(\rho_{\text{metal}}-\rho_{\text{fluid}})gR^2}{\mu} = \frac{2}{9}\cdot\frac{9000\cdot 10\cdot 10^{-4}}{1000} = \frac{2}{9}\cdot 9\times 10^{-3} = 2\times 10^{-3}\,\text{m/s}$. (Option (b) ignores buoyancy, using ρ_{metal} instead of $\rho_{\text{metal}} - \rho_{\text{fluid}}$; option (c) forgets the factor $\frac{2}{9}$; option (d) uses R instead of R^2.)

Question 27

A cantilever of free length $L = 0.1\,\text{m}$, circular cross-section of radius $r_0 = 1\,\text{mm}$, Young's modulus $E = 1\,\text{GPa}$, and density $\rho = 1000\,\text{kg/m}^3$ sags under its own weight ($g = 10\,\text{m/s}^2$). What is the Euler–Bernoulli tip deflection $\delta = qL^4/(8EI)$, with $q = \rho Ag$?

(a) 4.0 mm

(b) 0.25 mm

(c) 2.0 mm

(d) **0.5 mm** ✓

Answer: (d)
Solution. With $A = \pi r_0^2$ and $I = \pi r_0^4/4$: $\delta = \frac{\rho g L^4}{2Er_0^2} = \frac{1000\cdot 10\cdot 10^{-4}}{2\cdot 10^9\cdot 10^{-6}} = 5\times 10^{-4}\,\text{m} = 0.5\,\text{mm}$.

Question 28

In the small-deflection regime, the free length of the cantilever in Question 27 is doubled (all else unchanged). By what factor does the tip deflection change?

(a) 2

(b) 4

(c) **16** ✓

(d) 8

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Answer: (c)
Solution. $\delta = qL^4/(8EI)$, and q (weight per unit length) does not change, so $\delta \propto L^4$: doubling L multiplies δ by $2^4 = 16$.

Question 29

A horizontal cantilever of length L is clamped at one end and sags under its own weight (a uniform load q per unit length). Euler–Bernoulli beam theory predicts the tip deflection $\delta_{\text{EB}} = qL^4/(8EI)$, which grows linearly with q and can even exceed L for a large enough load. A DER simulation with a large load gives a vertical tip deflection much smaller than δ_{EB} (and always less than L). What is the main reason for the difference?

(a) **Euler–Bernoulli theory is linear: it assumes small slopes and a shape that stays close to the undeformed one. DER makes no such assumption.** ✓

(b) Implicit Euler adds artificial damping that reduces the DER deflection.

(c) The rod stretches a lot at large loads.

(d) DER includes inertia, while Euler–Bernoulli theory does not.

Answer: (a)
Solution. Euler–Bernoulli theory is geometrically linear (small slopes, fixed geometry), so its deflection is linear in the load and overpredicts large deflections (it can even predict a deflection larger than L). DER is geometrically nonlinear: as the rod rotates toward the vertical, the moment arm of the weight shrinks and the vertical deflection saturates below L. At steady state, damping and inertia do not affect the shape, and stretching is negligible.

Question 30

In DER, the bending energy at node k is $\frac{1}{2}EI\,|\boldsymbol{\kappa}_k - \bar{\boldsymbol{\kappa}}_k|^2/\bar{l}_k$, where $\boldsymbol{\kappa}_k$ is the curvature and $\bar{\boldsymbol{\kappa}}_k$ is the natural (stress-free) curvature, and $\bar{l}_k$ is the undeformed Voronoi length of node k. A rod clamped at one end is initialized in the shape of a circular arc, and there is no gravity or other load. Normally $\bar{\boldsymbol{\kappa}}$ is computed from this initial shape, which makes the rod naturally curved. Instead, a student sets $\bar{\boldsymbol{\kappa}}_k = \mathbf{0}$ at every node (a naturally straight rod) but still starts the simulation from the circular arc. What happens?

(a) It keeps its curved shape, because there is no load.

(b) **It opens up into a nearly straight rod, because the bending energy is zero only when the rod is straight.** ✓

(c) It curls up more tightly.

(d) The simulation fails, because the natural curvature must not be zero.

Answer: (b)
Solution. The bending energy $\frac{1}{2}EI|\boldsymbol{\kappa}_k - \bar{\boldsymbol{\kappa}}_k|^2/\bar{l}_k$ is zero only when $\boldsymbol{\kappa}_k = \bar{\boldsymbol{\kappa}}_k$. With $\bar{\boldsymbol{\kappa}} = \mathbf{0}$, the stress-free shape is straight, so the rod, started from the arc, opens up into a nearly straight rod (HW3, Question 11).

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Question 31

Use x, y, z coordinates with gravity along $-z$. The two rods described below are both clamped at one end and simulated with DER. Rod A is naturally straight and initially lies along the x-axis. Rod B is naturally curved: its stress-free shape is a circular arc in the x–y (horizontal) plane, and it starts in that shape. Both rods sag under gravity. Which statement about their twist angles is true?

(a) Both rods twist, because gravity always twists a rod.

(b) Neither rod twists, because the first twist angle is fixed.

(c) **Rod A does not twist (it bends only in the x–z plane that contains it); rod B twists (gravity bends it out of the x–y plane in which it is curved).** ✓

(d) Rod A twists, but rod B does not.

Answer: (c)
Solution. Rod A bends within the vertical x–z plane that contains it, so its twist angles stay zero. Rod B is curved in the horizontal x–y plane and gravity pulls it out of that plane (along $-z$), which couples bending and twisting, so it develops nonzero twist (HW3, Question 10).

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